Taylor and Maclaurin Series

ANSWER KEY

Section 1: Common Maclaurin Series (e^x, $\sin x$, $\cos x$)

1 What is the general term in the Maclaurin series for $\sin x$?

a. $\frac{(-1)^n x^{2n+1}}{(2n+1)!}$ ✓

b. $\frac{(-1)^n x^{2n}}{(2n)!}$

c. $\frac{x^n}{n!}$

d. $\frac{x^{2n+1}}{(2n+1)!}$

The Maclaurin expansion for $\sin x$ contains only odd powers with alternating signs: $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots$

2 What is the coefficient of x^4 in the Maclaurin series for e^{-2x}?

a. $\frac{2}{3}$ ✓

b. $\frac{16}{24} = \frac{2}{3}$

c. $-\frac{2}{3}$

d. $\frac{16}{4}$

General term for e^u is $\frac{u^n}{n!}$. For $u = -2x$ and $n = 4$: $\frac{(-2x)^4}{4!} = \frac{16x^4}{24} = \frac{2}{3}x^4$.

3 Which series equals $\sum_{n=0}^{\infty} \frac{(-1)^n \pi^{2n}}{(2n)!}$?

a. $\cos(\pi) = -1$ ✓

b. $\sin(\pi) = 0$

c. e^{π}

d. 1

This is the Maclaurin series for $\cos x$ evaluated at $x = \pi$, which is $\cos(\pi) = -1$.

Section 2: Constructing Taylor Polynomials from Derivatives

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4 Which formula gives the n-th degree Taylor polynomial $T_n(x)$ centered at $x = c$?

a. $\sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x-c)^k$ ✓

b. $\sum_{k=0}^{n} f^{(k)}(c)(x-c)^k$

c. $\sum_{k=0}^{n} \frac{f(c)}{k!}(x-c)^k$

d. $\frac{f^{(n)}(c)}{n!}(x-c)^n$

By definition, the n-th degree Taylor polynomial is $T_n(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \ \dots\ + \frac{f^{(n)}(c)}{n!}(x-c)^n$.

5 Find the 2nd degree Taylor polynomial for $f(x) = \sqrt{x}$ centered at $c = 4$.

a. $2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2$ ✓

b. $2 + \frac{1}{4}(x-4) + \frac{1}{32}(x-4)^2$

c. $2 + \frac{1}{2}(x-4) - \frac{1}{8}(x-4)^2$

d. $2 + \frac{1}{4}(x-4)$

$f(4) = 2$. $f'(x) = \frac{1}{2\sqrt{x}} \Rightarrow f'(4) = \frac{1}{4}$. $f''(x) = -\frac{1}{4}x^{-3/2} \Rightarrow f''(4) = -\frac{1}{32}$. Coefficient $= \frac{-1/32}{2!} = -\frac{1}{64}$. $T_2(x) = 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2$.

6 If a function f has $f(2) = 5, f'(2) = -3, f''(2) = 8$, what is $T_2(2.1)$?

a. 4.74 ✓

b. 4.70

c. 4.82

d. 5.04

$T_2(x) = 5 - 3(x-2) + \frac{8}{2}(x-2)^2 = 5 - 3(x-2) + 4(x-2)^2$. At $x = 2.1$: $5 - 3(0.1) + 4(0.01) = 5 - 0.3 + 0.04 = 4.74$.

Section 3: Manipulating Known Taylor Series

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7 What is the first non-zero term of the Maclaurin series for $f(x) = x^2\cos(3x)$?

a. x^2 ✓

b. $3x^2$

c. $-x^2$

d. $\frac{9x^4}{2}$

$\cos(3x) = 1 - \frac{(3x)^2}{2!} + \ldots = 1 - \frac{9x^2}{2} + \ldots$ Multiplying by x^2: $x^2 - \frac{9x^4}{2} + \ldots$ First term is x^2.

8 Find the Maclaurin series for $\frac{\sin x - x}{x^3}$.

a. $-\frac{1}{6} + \frac{x^2}{120} - \ldots$ ✓

b. $\frac{1}{6} - \frac{x^2}{120} + \ldots$

c. $1 - \frac{x^2}{6} + \ldots$

d. $-\frac{x^3}{6} + \ldots$

$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \ldots \Rightarrow \sin x - x = -\frac{x^3}{6} + \frac{x^5}{120} - \ldots$ Dividing by x^3 yields $-\frac{1}{6} + \frac{x^2}{120} - \ldots$

9 Evaluate $\lim_{x\to 0} \frac{\cos x - 1 + \frac{x^2}{2}}{x^4}$.

a. $\frac{1}{24}$ ✓

b. $-\frac{1}{24}$

c. 0

d. $\frac{1}{12}$

$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots \Rightarrow \cos x - 1 + \frac{x^2}{2} = \frac{x^4}{24} - \ldots$ Dividing by x^4 gives $\frac{1}{24}$.

Section 4: Lagrange Error Bound (Taylor's Remainder Theorem)

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10 What is the Lagrange Error Bound formula for $|R_n(x)| = |f(x) - P_n(x)|$ centered at c?

a. $|R_n(x)| \le \frac{M}{(n+1)!}|x-c|^{n+1}$, **where** $M = \max|f^{(n+1)}(t)|$ ✓

b. $|R_n(x)| \le \frac{M}{n!}|x-c|^n$

c. $|R_n(x)| \le |f^{(n+1)}(c)|$

d. $|R_n(x)| \le \frac{1}{(n+1)!}$

Taylor's Remainder Theorem states $|R_n(x)| \le \frac{M}{(n+1)!}|x-c|^{n+1}$ where M is the maximum value of $|f^{(n+1)}(t)|$ on the interval between c and x.

11 If $P_3(x)$ approximates $\sin x$ at $x = 0.5$ centered at 0, what is the Lagrange Error Bound using $M = 1$?

a. $\frac{(0.5)^4}{24} = \frac{1}{384}$ ✓

b. $\frac{(0.5)^3}{6}$

c. $\frac{1}{24}$

d. $\frac{(0.5)^4}{120}$

For $n = 3$, $n+1 = 4$. $M = \max|\sin^{(4)}(t)| = \max|\sin t| \le 1$. Bound $= \frac{1}{4!}(0.5)^4 = \frac{0.0625}{24} = \frac{1}{384} \approx 0.0026$.

12 On an AP free-response question, which derivative is bounded to establish the Lagrange error bound for a 4th-degree Taylor polynomial?

a. **The 5th derivative** $f^{(5)}(x)$ ✓

b. The 4th derivative $f^{(4)}(x)$

c. The 1st derivative $f'(x)$

d. The 6th derivative $f^{(6)}(x)$

For an n-th degree Taylor polynomial, the remainder formula uses the $(n+1)$-th derivative. For $n = 4$, this is the 5th derivative.

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